

MCT-454L – Mobile Robotics

Lab-02

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1. Objective

The purpose of this lab was to gain hands-on experience with ROS 2 tools and understand how different components such as nodes, topics, and services interact within a robotic system. Another key objective was to control and monitor a simulated robot using both command-line tools and graphical interfaces.

2. Procedure and Implementation

The lab was performed step-by-step using the Turtlesim simulator to explore ROS 2 functionalities.

a) Environment Setup

The ROS 2 environment was initialized in the terminal to enable access to ROS commands and packages. The turtlesim package was also verified to ensure the simulation could be executed properly.

b) Launching the Simulation

The Turtlesim node was executed, which opened a graphical window displaying a turtle inside a bounded environment. This node acts as a publisher and subscriber, forming the basis for interaction in the simulation.

c) Manual Control of the Robot

A separate node was launched to control the turtle using keyboard inputs. This teleoperation node published velocity commands to the turtle, allowing real-time movement. The interaction demonstrated how one node can control another through topic communication.

d) Exploring Topics

Active topics in the system were listed to understand communication channels. The pose topic of the turtle was monitored, which continuously published position and orientation data. Observing this stream helped in understanding how sensor-like data is shared in ROS 2.

e) Publishing Velocity Commands

Instead of manual control, velocity commands were directly published to the turtle using the command line. By specifying linear and angular velocities, the turtle's movement could be controlled programmatically. This demonstrated how robots can be automated using topic-based communication.

f) Using Services

A service call was made to reset the simulation environment. Unlike topics, services follow a request-response model. This step highlighted the difference between continuous communication (topics) and one-time actions (services).

3. Working with rqt GUI

The rqt graphical tool was used to visualize and interact with the ROS 2 system more intuitively.

a) System Visualization

Through rqt plugins, all active nodes, topics, and services were explored. This provided a clear understanding of how different components were interconnected in the system.

b) Calling Services via GUI

Services such as reset were triggered directly from the GUI. Additionally, a new turtle was spawned by providing position and orientation parameters. This showed how services can dynamically modify the simulation.

c) Controlling Multiple Robots

After spawning a second turtle, it was controlled independently by publishing commands to its respective topic. This demonstrated the scalability of ROS 2 systems where multiple entities can operate simultaneously without interference.

d) Experimentation with Additional Services

Different services were tested, including changing the background color and teleporting the turtle. These experiments reinforced the understanding of how services can alter the system state instantly.

4. Observations and Analysis

- ROS 2 follows a modular architecture where nodes communicate through well-defined interfaces.
- Topics are effective for continuous data streaming such as position updates.
- Services are suitable for discrete actions like resetting or spawning objects.
- The “rqt” tool simplifies debugging and system visualization significantly.
- Multiple robots can be controlled independently by assigning separate namespaces and topics.

5. Issues Encountered

- Initially, ROS commands did not work due to the environment not being sourced. This was resolved by properly sourcing the setup file.
- While spawning a second turtle, incorrect parameter values caused failure, which was corrected by providing valid inputs.

6. Conclusion

This lab provided practical exposure to ROS 2 concepts by integrating command-line tools with a graphical simulator. It helped in understanding how nodes, topics, and services function together in a robotic system. The use of both CLI and rqt enhanced comprehension of system behavior and control mechanisms. Overall, the lab successfully met its objectives and strengthened foundational knowledge required for more advanced robotics applications.